

AI/ML Internship – Day 22 Notes & Tasks

Module 4: NumPy (Aggregations & Statistics)

Part 1: Why Aggregation is Important?

 In AI/ML:

- We summarize data
- We extract insights
- We understand patterns

 Example:

- Average marks
 - Highest sales
 - Minimum temperature
-

Part 2: Theory Answers (IMPORTANT)

1. What are Aggregation Functions?

 Aggregation functions are used to **combine multiple values into a single value**.

 Example:

- sum → total
 - mean → average
-

2. What is Mean?

 Mean is the **average value** of data.

 Formula:

Mean = Sum of values / Number of values

3. What is Median?

👉 Median is the **middle value** after sorting data.

👉 Example:

- [1, 2, 3] → median = 2
 - [1, 2, 3, 4] → median = (2+3)/2
-

◆ 4. What is Standard Deviation?

👉 It measures how much data **varies from the mean**.

👉 Low value → data is close

👉 High value → data is spread

◆ 5. What is Axis?

👉 Axis defines **direction of operation** in arrays.

- axis=0 → column-wise
 - axis=1 → row-wise
-

📌 Part 3: Basic Aggregation Functions

```
import numpy as np
```

```
arr = np.array([10, 20, 30, 40])
```

```
print(np.sum(arr))
```

```
print(np.mean(arr))
```

```
print(np.max(arr))
```

```
print(np.min(arr))
```

📌 Part 4: More Statistical Functions

```
print(np.median(arr))
```

```
print(np.std(arr))
```

```
print(np.var(arr))
```

◆ Explanation

- median → middle value
 - std → spread of data
 - var → variance
-

📌 Part 5: Aggregation in 2D Arrays

```
arr = np.array([
    [1, 2, 3],
    [4, 5, 6]
])
```

◆ Column-wise

```
print(np.sum(arr, axis=0))
```

👉 Output:

```
[5 7 9]
```

◆ Row-wise

```
print(np.sum(arr, axis=1))
```

👉 Output:

```
[6 15]
```

📌 Part 6: Min & Max Index

```
print(np.argmax(arr))
print(np.argmin(arr))
```

👉 Returns index positions

📌 Part 7: Cumulative Operations

```
arr = np.array([1, 2, 3, 4])
```

```
print(np.cumsum(arr))
```

👉 Output:

```
[1 3 6 10]
```

Part 8: Real-World Thinking

👉 In AI:

- Mean → average prediction
- Max → best value
- Std → data consistency
- Axis → feature-wise analysis

👉 Used in:

- Data analysis
 - Feature engineering
 - Model evaluation
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Tasks for Day 22

Task 1: Theory (Write Answers)

1. What are aggregation functions?
 2. What is mean?
 3. What is median?
 4. What is standard deviation?
 5. What is axis?
-

Task 2: Basic Aggregation

1. Find sum

2. Find mean
 3. Find max and min
-

Task 3: Statistical Functions

1. Find median
 2. Find standard deviation
 3. Find variance
-

Task 4: Axis Practice

1. Row-wise sum
 2. Column-wise sum
 3. Mean using axis
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Task 5: Index Functions

1. Find index of max
 2. Find index of min
-

Task 6: Cumulative

1. Compute cumulative sum
2. Analyze output